

Bridging Graph Polynomials and Symmetric Tensors

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Overview

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 - 2 Establish a Belkale-Brosnan type result for symmetric tensors

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- Notation: $V^{\otimes d}$ denotes the tensor product of a vector space V with itself d times. $S^d(V) \subset V^{\otimes d}$ denotes the subspace containing all the symmetric tensors.
- Elementary symmetric tensors are of the form $v^{\otimes d}$.

Relationship with Homogenous Polynomials

Consider $k[x_1 \dots x_n]_d$, the set of all homogenous polynomials of degree d over k (implicitly included in this is the 0 polynomial). Then an established result is that

$$k[x_1 \dots x_n]_d \cong S^d(k^n)$$

as k -vector spaces, by linearly extending the mapping of monomials

$$\mathbf{x}^\alpha := \prod_{i=1}^n x_i^{a_i} \mapsto \frac{1}{d!} \sum_{\sigma \in S_d} \sigma \left(\bigotimes_{i=1}^n e_i^{\otimes a_i} \right)$$

Here α denotes the tuple $(a_1, a_2, \dots, a_n)$ where $a_i \geq 0 \forall i = 1, 2, \dots, n$, and for this mapping $|\alpha| = \sum_{i=1}^n a_i = d$.

Veronese Variety

Geometrically, consider $\mathbb{P}V$ to be the set of all lines in $V \cong k^n$ passing through the origin. Then the Veronese mapping given by

$$\nu_d : \mathbb{P}V \rightarrow \mathbb{P}k^N$$

taking points

$$[x_1 : x_2 : \dots : x_n] \mapsto [x_1^d : x_1^{d-1}x_2 : \dots : x_n^d]$$

allows for the visualisation of elementary (rank-1) symmetric tensors as points on a variety in $\mathbb{P}k^N$. Here $N = \binom{n-1+d}{d}$. More general symmetric tensors then lie on secant varieties to the Veronese.

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 $b_0(G) = 1, b_1(G) = 0$.
- A *spanning tree* $T \leq G$ contains all of V but not (necessarily) all of E .

Constructing (particular) Graph Polynomials

For the rest of this presentation $G = (V, E)$ such that $\|G\| = n$ and $b_0(G) = 1$. A *Kirchoff* polynomial for a given graph G is given by

$$P_G(x_1, \dots, x_n) := \sum_{T \leq G} \prod_{e \notin T} x_e$$

Note the variables of the polynomial correspond to a choice of labelling the edges.

Properties of the Kirchoff Polynomial

- They are homogenous polynomials, and lie in the quotient ring $R := k[x_1, x_2, \dots, x_n]/(x_1^2, x_2^2, \dots, x_n^2)$

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- The degree of each Kirchoff polynomial is exactly $b_1(G)$
- Variables common to each term in the polynomial denote self-loops of G .
- An equivalence relation $\sim_S$ can be imposed by $P_A \sim_S P_B \iff P_A(x_1, \dots, x_n) = \tau \cdot P_B(x_1, \dots, x_n)$. Here $\tau \in S_n$ permutes n elements, corresponding to a relabelling of edges.

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 $[X] = \{V \mid \pi : X \dashrightarrow V \text{ a birational morphism exists}\}$; modulo all relations of the form $[X] = [X \setminus Y] + [Y]$. Addition here is defined as $[X] + [Y] = [X \sqcup Y]$

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- This group can be seen as a commutative ring with unity by defining multiplication $[X][Y] = [X \times_k Y]$.
- $\mathcal{K}(V_k)$ can be seen as a module over $\mathbb{Z}[\mathbb{L}]$ where $\mathbb{L} := [\mathbb{A}^1]$

Belkale, Brosnan (2004)

Let $\mathbb{V}(P_G) \subset \mathbb{A}^n$ denote the scheme of zeroes of P_G . Then define $Y_G = \mathbb{A}^n \setminus \mathbb{V}(P_G)$. Then

Theorem (Belkale, Brosnan 2004)

Define $S := \{\mathbb{L}^n - \mathbb{L}\}_{n \geq 2}$ as a multiplicative subset in $\mathcal{K}(V_k)$. Then the collection of $[Y_G]$ generate $S^{-1}\mathcal{K}(V_k)$ as a $S^{-1}\mathbb{Z}[\mathbb{L}]$ module.

Aluffi, Marcolli 2011

X, Y are stably birational iff $X \times \mathbb{P}k^m \dashrightarrow Y \times \mathbb{P}k^n$ is birational, for $n, m \geq 0$.

Theorem (Larsen, Lunts (2001))

$\mathcal{K}(V_k)/(\mathbb{L}) \cong \mathbb{Z}[SB]$, the ring of stably birational varieties

Theorem (Aluffi, Marcolli (2011))

The images of $[Y_G]$ in $\mathcal{K}(V_k)$ generate $\mathbb{Z} \subsetneq \mathbb{Z}[SB]$

Corollary

Suppose $S' := \{\mathbb{L}^n - 1\}_{n \geq 1}$. Then $[Y_G]$ do not generate $S'^{-1}\mathcal{K}(V_k)$ over $S'^{-1}\mathbb{Z}[\mathbb{L}]$

Graphs as a Disjoint Union

Let $\mathcal{G}_n$ denote the family of all connected graphs with a fixed number of edges n .

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- $\mathbf{G}_n$ can then be expressed as the disjoint union $\bigsqcup_{i=0}^n \mathbf{G}_{n,i}$ where each component contains the graphs with $b_1(G) = i$.

Addition in Graphs?

In $\mathbf{G}_n$, an attempt to define an addition operation is given below

- Let $[A]_{KP,S}, [B]_{KP,S} \in \mathbf{G}_n := (\mathcal{G}_n / \sim_{KP}) / \sim_S$

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- Choosing P_C as an element with the largest number of summands in this set. $[P_C]_S$ then is a corresponding equivalence class.
- **Conjecture:** Does there always exist C a graph such that its Kirchoff polynomial is P_C ? If yes then
- This turns $\mathbf{G}_n$ into an additive abelian group

Image of R in symmetric tensors

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Localisations in Context

Generating of Symmetric Tensors

Further Questions

Bibliography